

SDF Laboratory Assignment Solutions & Manual

Course Code: 25B17CI172 • Computing Centre AB2 • JUIT Solan

1. Lab Environment & GCC Guidelines

- Linux Terminal GCC Compilation: `gcc -Wall -Wextra lab1.c -o lab1`
- Execution: `./lab1`
- Standard Header Files: `<stdio.h>`, `<stdlib.h>`, `<string.h>`, `<math.h>`

2. Weekly Lab Syllabus (Exercises 1-12)

- Lab 1: Basic Input/Output, Data Types, and Arithmetic Operations.
- Lab 2: Conditional Statements, switch-case, and Quadrant Calculators.
- Lab 3: Loops, Prime Number Check, Fibonacci Sequence, Series Sum.
- Lab 4: 1D Arrays, Linear Search, Bubble Sort, Min/Max Finders.
- Lab 5: 2D Matrix Multiplication, Transpose, and Symmetry Checks.
- Lab 6: Strings, Palindrome, Tokenization, and Vowel Counter.
- Lab 7: Modular Programming with Functions and Recursion (Tower of Hanoi).
- Lab 8: Pointers, Call by Value vs Reference, Pointer Arithmetic.
- Lab 9: Dynamic Memory Allocation (malloc, calloc, realloc, free).
- Lab 10: Structures (struct), Array of Structures, Nested Structures.
- Lab 11: File Handling, Sequential Access, Word Counting.
- Lab 12: Final Mini-Project & Capstone Code Demonstration.